

Chapter 6

Electronic Structure of Atoms

1. The wavelength of light emitted from a traffic light having a frequency of 5.75×10^{14} Hz is _____.

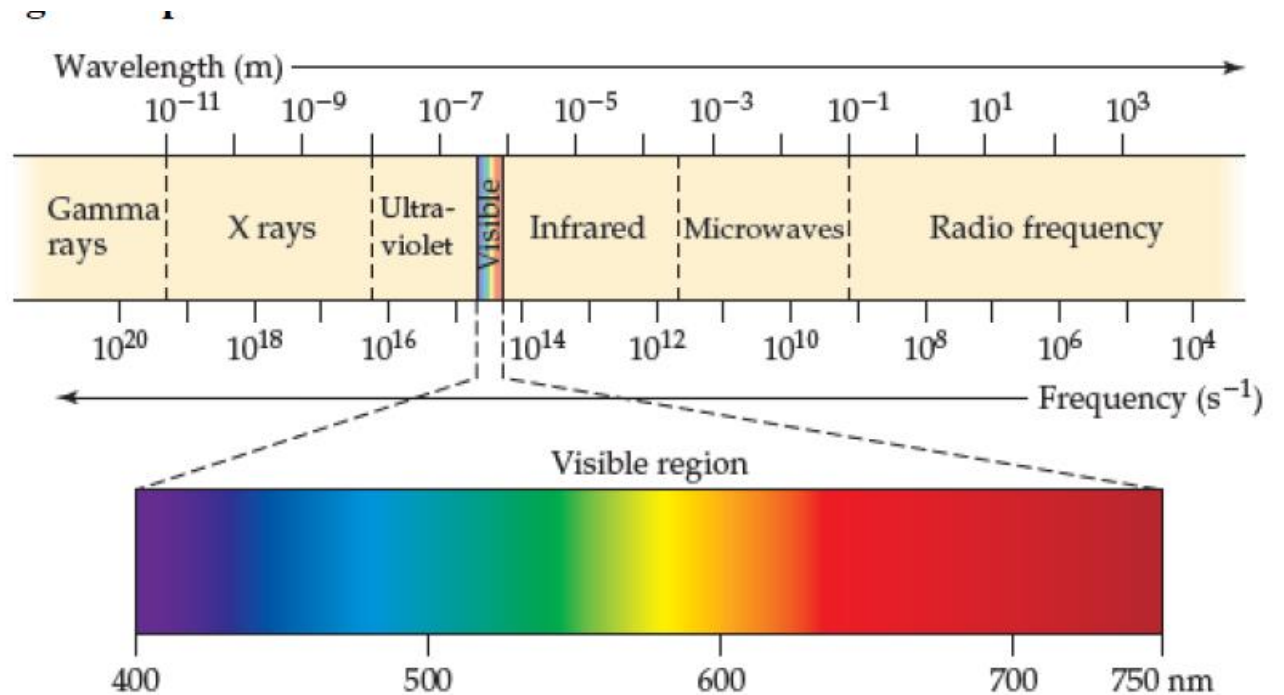
- A) 702 nm
- B) 641 nm
- C) 674 nm
- D) 522 nm**
- E) 583 nm



$$\lambda \nu = c$$

2. Of the following, _____ radiation has the shortest wavelength.

- A) X-ray
- B) radio
- C) microwave
- D) ultraviolet
- E) infrared



3. The photoelectric effect is _____.

- A) the total reflection of light by metals giving them their typical luster
- B) the production of current by silicon solar cells when exposed to sunlight
- ☒ C) the ejection of electrons by a metal when struck with light of sufficient energy
- D) the darkening of photographic film when exposed to an electric field
- E) a relativistic effect

4. Of the following transitions in the Bohr hydrogen atom, the B transition results in the emission of the highest-energy photon.

A) $n = 1 \rightarrow n = 6$

B) $n = 6 \rightarrow n = 1$

C) $n = 6 \rightarrow n = 3$

D) $n = 3 \rightarrow n = 6$

E) $n = 1 \rightarrow n = 4$

$$\begin{aligned} B) \quad & -2.18 \times 10^{-18} \text{ J} \cdot \left(\frac{1}{1^2} - \frac{1}{6^2} \right) \\ & = -2.12 \times 10^{-18} \text{ J} \end{aligned}$$

5. According to the Heisenberg Uncertainty Principle, it is impossible to know precisely both the position and the _____ of an electron.

- A) mass
- B) color
- ☒ C) momentum
- D) shape
- E) charge

Uncertainty principle: $\Delta x \cdot \Delta(\mathbf{mv}) \geq \frac{h}{4\pi}$

6. All of the orbitals in a given subshell have the same value of the D quantum number.

A) principal

B) angular momentum

C) magnetic

D) A and B

E) B and C

7. Which of the subshells below do not exist due to the constraints upon the angular momentum quantum number _____?

A) 2d $n=2 \rightarrow l=0,1$

B) 2s

C) 2p

D) all of the above

E) none of the above

8. An electron cannot have the quantum numbers $n =$ _____,
 $l =$ _____, $m_l =$ _____.

A) 6, 1, 0

B) 3, 2, 3

C) 3, 2, -2

D) 1, 0, 0

E) 3, 2, 1

$l=2$ then $m_l = -2, -1, 0, 1, 2$

The magnetic quantum number (m_l): orientation, $m_l =$ interval of $[-l, +l]$

9. Which quantum number determines the energy of an electron in a hydrogen atom _____?

A) n

B) E

C) m_l

D) l

E) n and l

number (n): energy level, $n = 1, 2, 3, 4, \dots$

quantum number (l): shape (of orbital), $l = 0, 1, 2, 3, \dots n-1$

number (m_l): orientation, $m_l = \text{interval of } [-l, +l]$

10. In a p_x orbital, the subscript x denotes the E of the electron.

- A) energy
- B) spin of the electrons
- C) probability of the shell
- D) size of the orbital
- ☒ E) axis along which the orbital is aligned

11. At maximum, an f -subshell can hold 14 electrons, a d -subshell can hold 10 electrons, and a p -subshell can hold 6 electrons.

A) 14, 10, 6

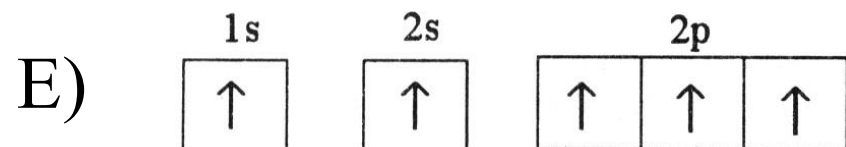
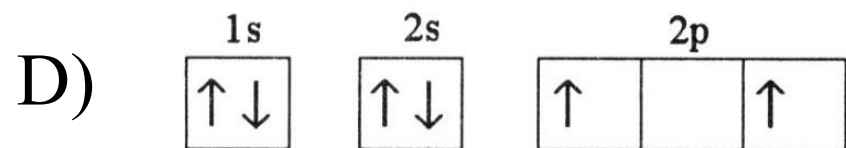
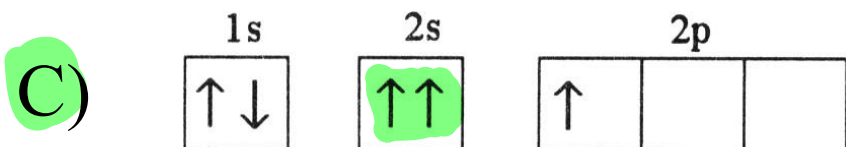
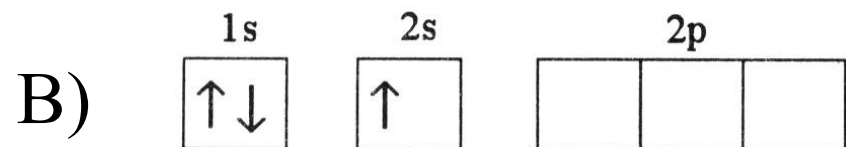
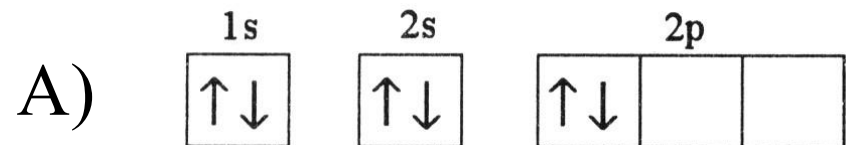
B) 2, 8, 18

C) 14, 8, 2

D) 2, 12, 21

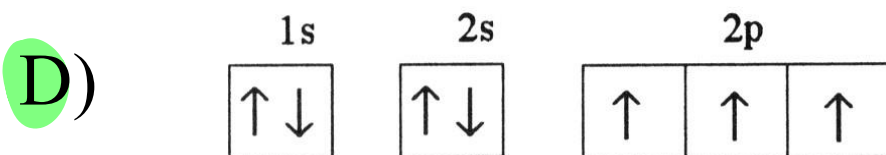
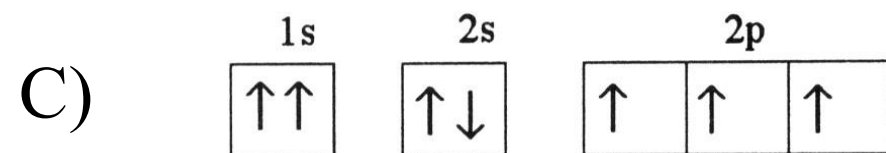
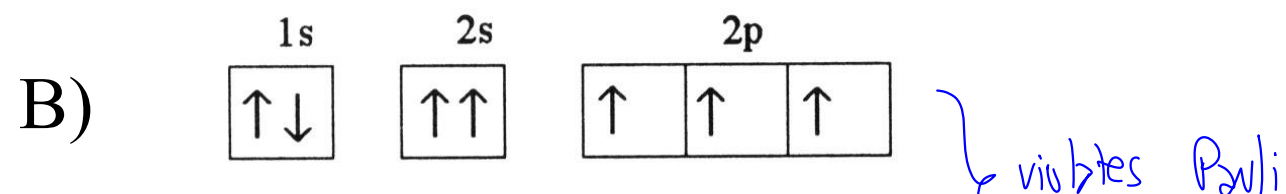
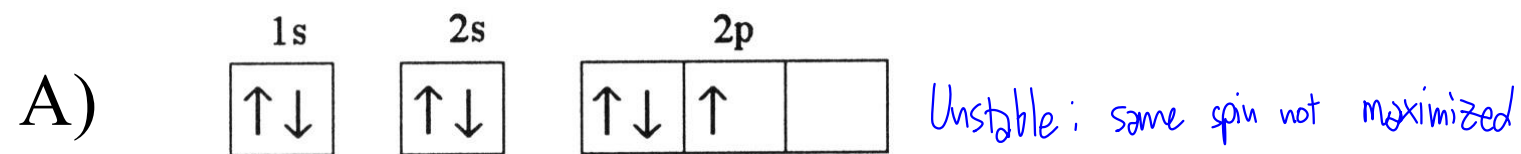
E) 2, 6, 10

12. Which electron configuration represents a violation of the Pauli exclusion principle _____ **C** _____?



The Pauli exclusion principle: an orbital can hold a maximum of two electrons and they must have opposite spins.

13. Which one of the following is the correct electron configuration for a ground-state nitrogen atom _____?



E) None of the above is correct.